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RESEARCH

# Synergistic sulfur engineering in Amorphous Co–S nanoparticles via ChCl–EG DES–mediated synthesis for efficient overall water splitting

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**Abstract**

This study presents a green strategy for synthesizing amorphous cobalt sulfide (Co–S) bifunctional electrocatalysts via a choline chloride–ethylene glycol deep eutectic solvent (DES) under ambient conditions, addressing ionic coordination dynamics and defect engineering for enhanced solid–state ionic/electronic transport in energy conversion. By modulating the equilibrium between $Co^{2+}$ and $S_2O_3^{2-}$ ions in Ethaline, we fabricated monodisperse Co–150S nanoparticles (~78 nm) with tailored sulfur content (S/Co = 1.9), an amorphous architecture, and abundant oxygen vacancies. These structural features synergistically optimized ionic diffusion pathways and electronic conductivity, achieving exceptional hydrogen evolution reaction (HER) and oxygen evolution reaction (OER) activities in alkaline media. The Co–150S/NF electrode demonstrated a volcano-like sulfur-dependent activity profile, achieving the highest electrochemically active surface area (ECSA, 3.7 $cm^2$) and ultralow overpotentials of − 105 mV (HER) and 277 mV (OER) at 10 mA $cm^{-2}$, comparable to benchmark Pt/C and $RuO_2$ catalysts. Post-electrolysis characterization revealed dynamic structural reorganization during HER and OER operations, involving over 80 at% sulfur depletion and the formation of metastable Co-rich phases that maintained catalytic functionality. In overall water splitting, the system required only 1.62 V to drive 10 mA $cm^{-2}$ with minimal activity decay (1.5 mV $h^{-1}$ over 26 h). Mechanistic investigations revealed that sulfur incorporation initiates a multiscale optimization process: (i) DES–mediated ionic confinement prevents particle aggregation, promoting uniform nanosphere formation; (ii) modulation of the electronic structure through nonstoichiometric Co–S coordination; and (iii) defect engineering through the enrichment of oxygen vacancies. This study provides insights into ionic coordination mechanisms in non–aqueous solvents and defect–mediated ion transport in amorphous solids, suggesting a potential strategy for developing electrocatalysts applicable to related energy storage technologies.

**Keywords** Deep eutectic solvent · Cobalt sulfide · Nano particles · Amorphous structure · Overall water splitting

## Introduction

In the global transition toward clean energy, water electrolysis for hydrogen production is widely regarded as a core pathway to achieving carbon neutrality due to its zero–carbon emission characteristics [1]. However, the sluggish kinetics of both the anode HER and the cathode OER lead to prohibitively high overpotentials ($\eta$), hindering large–scale commercialization of water electrolysis. To address these challenges, transition metal sulfides have emerged as compelling bifunctional electrocatalysts, offering an optimal combination of intrinsic activity, operational stability, and economic viability compared to noble metal counterparts [2, 3]. Among these materials, cobalt–based sulfides excel through synergistic

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features: (i) polymorphic structural adaptability, (ii) tunable electronic states, (iii) sulfur–rich coordination optimizing intermediate adsorption, and (iv) earth–abundance enabling scalability—collectively driving superior electrocatalytic water splitting performance [4–6]. Recent developments have produced diverse efficient cobalt sulfide bifunctional catalysts, such as $CoS_2$ [7–10], $MoS_2/Co_9S_8/Ni_3S_2/Ni$ [11], $Co_3S_4/CeO_2$ [12], $Co_4Ni_1S/CC$ [13], $NiCo_2S_4$ [14], Co–S–50/NF [15], $Co_9S_8@MoS_2$/CNFs [16], $CuNiCoS_4$/1 T–$MoS_2$ [17], $LaCoS_3$ [18], W–$Co_3S_4@Co_3O_4$ [19], FeCoNiS [20], $Ni_3S_2/Co_3S_4$ [21], Cu@$CoS_x$ [22]. These materials were typically synthesized through established methodologies including chemical vapor deposition [16], hydrothermal synthesis [11–14], solvothermal methods [7, 10], electrodeposition [15], high–temperature calcination [17–19], and templating [9, 20]. Interestingly, nearly all of these materials were engineered into nanostructures, including nanoparticles, nanowires, nanotubes, or nanosheets, which enhance the exposure of active sites and significantly improve their electrical conductivity and catalytic activity.

Beyond conventional synthetic approaches, deep eutectic solvent (DES)–mediated synthesis has emerged as an innovative green synthesis platform, capitalizing on self–assembling hydrogen–bond networks and dynamic metal coordination to enable precise nanostructuring under mild, ambient–pressure conditions. This method presents three distinct advantages: (i) Suppression of Ostwald ripening through high surface tension, low water activity, and ordered hydrogen bonding, enabling atomically precise assembly and morphology–controlled growth of nanomaterials at low temperatures; (ii) Facilitation of tailored vacancy concentrations via dynamic solvent–precursor interactions, optimizing hydrogen adsorption and oxygen intermediate binding; (iii) High recyclability (> 90%) of DES solvents with minimal toxic byproduct generation. Recent advancements in electrocatalyst design have highlighted the promise of DES–mediated synthesis strategies. Notably, Li [23] reported an Fe–based organic–inorganic hybrid electrocatalyst synthesized via DES, which achieved an OER overpotential of only 280 mV (Tafel slope: 47 mV $dec^{-1}$) in 1 M KOH—significantly outpacing the performance of commercial $IrO_2$ catalysts. Similarly, Liu et al. [24] demonstrated a hierarchically structured FeCoNi nitride–sulfide electrocatalyst prepared using DES, requiring just 245 mV overpotential for OER (Tafel slope: 38 mV $dec^{-1}$) in the same electrolyte. This performance far exceeded that of commercial $IrO_2$ (327 mV), further substantiating the viability of DES–driven synthetic strategies in developing high–efficiency electrocatalytic materials.

Herein, we report a green Ethaline/DES–assisted synthesis route for the low–temperature synthesis of amorphous cobalt sulfide (Co–S) bifunctional electrocatalysts under ambient pressure. By optimizing the dynamic coordination equilibrium between $Co^{2+}$ and $S_2O_3^{2-}$ in Ethaline (0.20 M $CoCl_2$, 0.15 M $Na_2S_2O_3$, 80 °C, 6 h), we fabricated monodisperse quasi–spherical Co–150S nanoparticles (Co–150S NPs, ~ 78 nm diameter). Comprehensive characterization revealed that this material featured an amorphous structure, high sulfur content (S/Co atomic ratio = 1.9), and abundant oxygen vacancies, which synergistically enhanced electronic conductivity and active site density. Electrochemical analysis demonstrated a volcano-type relationship between sulfur content and catalytic activity, with Co–150S/NF achieving the highest $C_{dl}$ (6.7 mF $cm^{-2}$) and ECSA (3.7 $cm^2$). In 1 M KOH, the Co–150S/NF electrode exhibited exceptional bifunctional catalytic performance, surpassing low–sulfur counterparts while achieving comparable performance to $RuO_2$ and Pt/C. Post-test characterization revealed dynamic structural evolution during prolonged HER and OER operation (28 h at 10 mA $cm^{-2}$), where sulfur depletion exceeding 80 atomic% (at%) triggered phase reorganization into vertically aligned cobalt-rich nanoflakes, contributing to the bifunctional excellent electrolysis activity and stability. During overall water splitting, the catalyst required only 1.62 V to drive a current density of 10 mA $cm^{-2}$ and demonstrated excellent long–term stability. Mechanistic studies reveal that sulfur incorporation triggers a multiscale optimization cascade: (1) morphological control leading to dispersed quasi–spherical nanostructures that suppress agglomeration, (2) electronic structure optimization via non–stoichiometric Co–S complexes that enhance conductivity, and (3) defect engineering via oxygen vacancy enrichment that lowers the reaction energy barrier.

## Experimental section

### Chemicals and reagents

All chemicals and solvents, including choline chloride (ChCl, $C_5H_{14}ClNO$, 99%), ethylene glycol (EG, $(CH_2OH)_2$, 99%), cobalt(II) chloride hexahydrate ($CoCl_2{\cdot}6H_2O$, ≥ 99%), sodium thiosulfate pentahydrate ($Na_2S_2O_3{\cdot}5H_2O$, ≥ 99.5%), potassium hydroxide (KOH, ≥ 95%), and deionized water (18.2 MΩ·cm), were purchased from Aladdin Reagent Co., Ltd. (China) and used without further purification. Commercial catalysts, including platinum on carbon (Pt/C, 20 wt% Pt), ruthenium(IV) oxide ($RuO_2$), and 5 wt% Nafion solution, were obtained from Sigma–Aldrich (USA). The deep eutectic solvent Ethaline was synthesized by magnetically stirring a mixture of ChCl and EG at a molar ratio of 1:2 (ChCl:EG) at 50 °C until a homogeneous transparent liquid was formed.

### Preparation of Co–S nanoparticle catalysts

Co–S nanoparticles (Co–S NPs) were synthesized through a one–step DES–assisted process in Ethaline. To investigate the effect of sulfur precursor concentration, a series of samples labeled as Co–xS (x = 50, 100, 150, 200, 250) were prepared, where 'x' denotes the molar concentration of $Na_2S_2O_3·5H_2O$ (50–250 mM). In a typical procedure, cobalt chloride hexahydrate ($CoCl_2·6H_2O$, 0.20 M) and sodium thiosulfate pentahydrate ($Na_2S_2O_3·5H_2O$, variable concentrations from 50 to 250 mM) were dissolved in 100 mL of Ethaline under magnetic stirring for 30 min. The solution was then transferred to a sealed quartz reactor and heated at 80 °C for 6 h under ambient pressure. The resulting black precipitate was collected by vacuum filtration, washed with deionized water and ethanol (three cycles each), and vacuum–dried at 60 °C for 8 h. For systematic parameter screening (Figures S1–S3), the sulfur precursor concentration was varied while fixing $CoCl_2$ at 0.20 M, reaction temperature at 80 °C, and duration at 6 h. The optimal condition (0.20 M $CoCl_2$, 150 mM $Na_2S_2O_3$) yielded Co–150S NPs with superior electrocatalytic activity. Prior to electrode fabrication, nickel foam (NF, 0.5 cm × 0.5 cm) was pretreated by degreasing in absolute ethanol for 15 min and ultrasonically cleaned in 5 vol.% HCl for 10 min to remove surface oxides, then rinsed with deionized water and dried at 60 °C. For electrode preparation, 1.0 mg of Co–S NPs (or commercial $RuO_2$/Pt/C) was dispersed in a mixture of 30 μL 5 wt.% Nafion solution, 470 μL ethanol, and 500 μL deionized water via 30 min ultrasonication to form a homogeneous ink. The ink (250 μL) was drop–casted onto the pretreated NF, achieving a catalyst loading of 1.0 mg $cm^{-2}$, and vacuum–dried at 60 °C for 12 h. Optimal synthesis conditions (0.20 M $CoCl_2$, 0.15 M $Na_2S_2O_3$, 80 °C, 6 h) were determined through systematic parameter screening (Figures S1–S3), yielding Co–150S NPs with superior electrocatalytic activity.

### Physical characterization

The morphology and elemental composition of the samples were characterized using field–emission scanning electron microscopy (FESEM, JSM–7800F, JEOL, Japan) equipped with an energy–dispersive X–ray spectrometer (EDS, X–Max100, Oxford Instruments, UK) at an accelerating voltage of 15 kV. X–ray diffraction (XRD) patterns were recorded on a SmartLab–9 X–ray diffractometer (Rigaku, Japan) with Cu Kα radiation (λ = 1.54 Å, 45 kV/200 mA) over a 2$\theta$ range of 10°–90° at a scanning rate of 5° $min^{-1}$. Transmission electron microscopy (TEM) and high–resolution TEM (HRTEM) analyses were performed on a Tecnai G2 F30 microscope (FEI, Netherlands) operated at 200 kV. The chemical states of elements were investigated by X–ray photoelectron spectroscopy (XPS, PHI 550 Versa Probe, Ulvac–PHI, Japan) using monochromatic Al Kα radiation (h$\nu$ = 1486.6 eV). All binding energies were calibrated against the adventitious carbon C 1 s peak at 284.8 eV. Ultraviolet–visible (UV–Vis) absorption spectra of Ethaline/DES systems were acquired on a U–5100 spectrophotometer (Hitachi, Japan) at room temperature, and Fourier–transform infrared (FT–IR) spectra were collected using a Bruker IFS–66 spectrometer (Germany) in the range of 400–4000 $cm^{-1}$ with a resolution of 4 $cm^{-1}$.

### Electrochemical analysis

All electrochemical measurements were performed using a PARSTAT 4000 electrochemical workstation (Ametek, USA) in a standard three–electrode system. The working electrode was Co–xS/NF (0.5 cm × 0.5 cm), with a Hg/HgO reference electrode (1.0 M KOH) and a graphite rod counter electrode (diameter: 6 mm). The electrolyte was 1.0 M KOH (pH 13.9) saturated with nitrogen for 30 min prior to testing. Linear sweep voltammetry (LSV) was conducted at a scan rate of 5 mV $s^{-1}$, with current densities normalized to the geometric area. Electrochemical impedance spectroscopy (EIS) was measured from $10^5$ to $10^{-2}$ Hz with a 5 mV amplitude at selected potentials, and the Nyquist plots were fitted using a Randles equivalent circuit to extract solution resistance ($R_s$) and charge transfer resistance ($R_{ct}$) (Table S1). All polarization curves for HER and OER were compensated for 95% iR drop based on $R_s$ values from EIS. Potentials were converted to the reversible hydrogen electrode (RHE) scale using the equation: $E_{RHE} = E_{Hg/HgO} + 0.0591 \times pH + 0.096$ V (at 25 °C). Catalyst stability was evaluated by cyclic voltammetry (CV, 100 mV $s^{-1}$), multi–step chronopotentiometry, and chronoamperometry at a fixed current density. The Faradaic efficiency was determined through a drainage experiment. To evaluate the electrochemically active surface area (ECSA), double–layer capacitance ($C_{dl}$) was determined by cyclic voltammetry within the non–faradaic potential window (0.93–1.03 V vs. RHE) at scan rates of 5–30 mV $s^{-1}$. The charging current density ($\Delta j = (j_{anodic} - j_{cathodic})/2$) was plotted at 0.98 V vs. RHE against scan rate, and $C_{dl}$ was calculated from the linear slope. ECSA was derived using the relationship: ECSA = $(C_{dl}/C_s) \times S$, where $C_s$ represents the specific capacitance of the bare NF substrate, and $S$ is the geometric electrode area.

## Results and discussion

Ethaline, a deep eutectic solvent (DES), was synthesized by mixing choline chloride (ChCl) and ethylene glycol (EG) at a 2:1 molar ratio, yielding a homogeneous, transparent liquid (Fig. 1a). Introducing 0.15 M $Na_2S_2O_3$ induced a milky–white

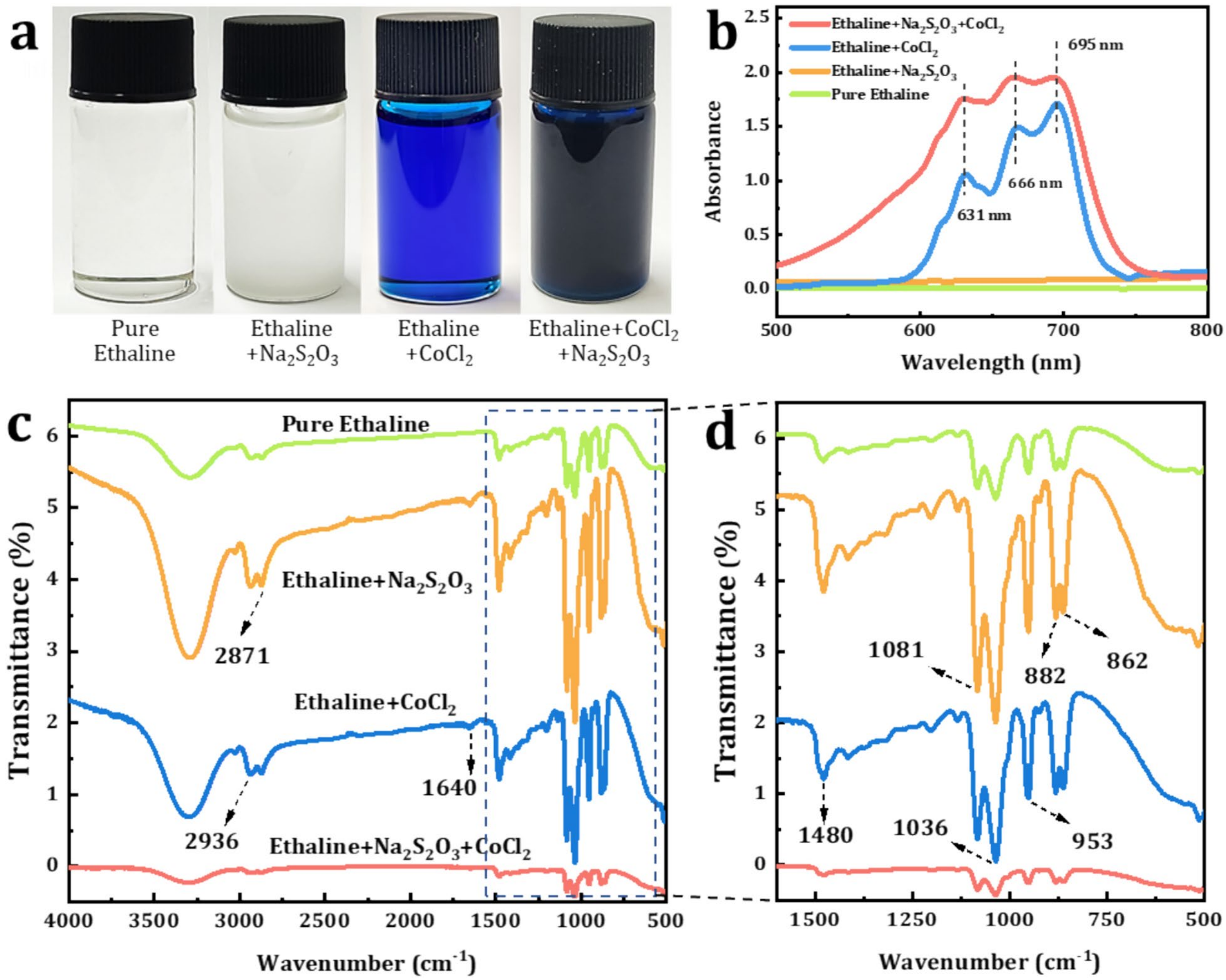


**Fig. 1** (**a**) Photographs, (**b**) UV–Vis spectra, (**c**) FT–IR spectra, and (**d**) enlarged view of the selected region in (**c**) for blank Ethaline, single–solute systems of 0.20 M $CoCl_2$–Ethaline and 0.15 M $Na_2S_2O_3$–Ethaline, and a binary system of 0.20 M $CoCl_2$–0.15 M $Na_2S_2O_3$–Ethaline

appearance, likely due to microdomain aggregation of thiosulfate anions. Subsequent addition of 0.20 M $CoCl_2$ resulted in a bright blue transparent solution, consistent with the characteristic color change of the $[CoCl_4]^{2-}$ complex [25]. When $CoCl_2$ and $Na_2S_2O_3$ coexisted in the system, the solution color shifted to dark blue, suggesting that the introduction of sulfur species altered the optical properties of the system.

UV–Vis spectroscopy (Fig. 1b) of the $CoCl_2$–Ethaline system revealed three characteristic absorption peaks at 631, 666, and 695 nm, which corresponded to the d–d electronic transitions of the tetrahedral $[CoCl_4]^{2-}$ complex [26, 27]. In the $CoCl_2$–$Na_2S_2O_3$–Ethaline ternary system, the peak positions remained unchanged, but the absorption intensity increased (Figs. 1c–d), indicating that the introduction of $S_2O_3^{2-}$ does not alter the tetrahedral coordination of $Co^{2+}$. However, it may have enhanced the structural stability of the $Co^{2+}$–Cl ligand through synergistic effects, or increased the oscillator strength of the d–d electronic transition, thereby influencing the optical characteristics of the system. FT–IR analysis (Fig. 1c and d) elucidated the hydrogen–bonding network of Ethaline. A broad absorption band in the 3500–3000 $cm^{-1}$ region is attributed to the coupled hydrogen–bonded vibrations of C–H, N–H, and O–H groups in the ChCl–EG system, which forms the three–dimensional framework structure of the DES [28, 29]. Additionally, characteristic peaks at 2936 and 2871 $cm^{-1}$ are assigned to the C–H stretching vibrations of methylene groups in EG, while the peak at 882 $cm^{-1}$ corresponds to the C–C skeletal vibrations in EG [30, 31]. The characteristic peaks of ChCl appear at 1480 $cm^{-1}$ for C–H in–plane bending vibrations, 1036 and 1081 $cm^{-1}$ for C–O asymmetric stretching vibrations, 1640 and 953 $cm^{-1}$ for C–C–O coupling vibrations, and 862 $cm^{-1}$ for C–H stretching vibrations [32–35]. Notably, the incorporation of $Co^{2+}$ and $S_2O_3^{2-}$ ions did not alter the characteristic peak positions or intensities (Figure S4), indicating that the DES's intrinsic hydrogen–bonding network remains intact despite the presence of exogenous species. Although localized perturbations in the molecular environment may arise from ion coordination or electrostatic effects, the hydrogen–bonded matrix exhibits dynamic reorganization capacity, thereby preserving its core vibrational signatures. These findings conclusively demonstrate the DES's structural robustness against external perturbations, establishing it as a stable and tunable microenvironment for precision nanoparticle synthesis [36, 37].

Figure 2 schematically illustrates the one–step Ethaline–assisted synthesis pathway of Co–S nanoparticles in the 0.20 M $CoCl_2$–0.15 M $Na_2S_2O_3$–Ethaline system. Within the ChCl–EG DES, $Co^{2+}$ ions preferentially coordinate with $Cl^-$ to form thermodynamically stable $[CoCl_4]^{2-}$ tetrahedral complexes [38, 39]. Meanwhile, $S_2O_3^{2-}$ anions accumulate at the surface of the $[CoCl_4]^{2-}$ complex through hydrogen bonding and electrostatic interactions, forming a precursor complex [40–42]. Under mild reaction conditions mediated by Ethaline (80 °C, 1 atm), the adsorbed $S_2O_3^{2-}$ undergoes proton–assisted disproportionation (Eq. 1), generating reactive $S^{2-}$ species that subsequently trigger interfacial ion exchange with $[CoCl_4]^{2-}$ via ligand substitution (Eq. 2). This stepwise process initiates Co–S nucleation through a coordinated dissolution–reconstruction mechanism:

$$S_2O_3^{2-} + H_2O \rightarrow S^{2-} + SO_4^{2-} + 2H^+ \quad (1)$$

$$[CoCl_4]^{2-} + S^{2-} \rightarrow Co-S \downarrow + 4Cl^- \quad (2)$$

The DES medium enables the controlled growth of nanoparticles through a triple regulation mechanism: (i) Spatial steric hindrance induced by the dynamic hydrogen–bond network suppresses particle aggregation; (ii) High viscosity retards ionic diffusion, modulating nucleation kinetics; (iii) Competitive coordination between $Cl^-$ and $S^{2-}$ balances crystalline facet development. This "coordination–dissociation–reconstruction" hierarchical reaction pathway, in synergy with the confinement effect of the DES, results in well–dispersed Co–S nanomaterials.

To elucidate the mechanism by which Ethaline–mediated synthesis parameters dictate product formation and catalytic performance, the interdependencies among reaction time, temperature, and sulfide precursor concentration via controlled experiments were systematically investigated. Optimal HER and OER catalytic performance were achieved at 80 °C with 0.15 M $Na_2S_2O_3$ over 6 h. As shown in Figures S1–S3, Co–S nanomaterials synthesized under these optimized conditions (80 °C, 0.15 M $Na_2S_2O_3$, 6 h) exhibited optimal HER and OER activities. The sulfur precursor concentration critically governed the reaction dynamics. An optimized $Na_2S_2O_3$ concentration modulated nucleation kinetics and growth pathways of active intermediates, which directly dictated the final product's morphology, stoichiometry, and crystallographic features. These structural parameters synergistically enhanced catalytic performance by tailoring active–site accessibility and electronic configurations (Figure S1). The inherent viscosity of Ethaline necessitated balanced thermal activation. A reaction temperature of 80 °C optimally enhanced mass transport while ensuring alignment with nucleation–growth kinetics (Figure S2). Reaction duration critically determined structural evolution. Insufficient reaction time (< 6 h) yielded incomplete activation, whereas prolonged duration (> 6 h) promoted Ostwald ripening and sulfur over–incorporation, both of which degraded performance via active–site depletion or structural collapse (Figure S3).

Having established the critical role of synthesis parameters in catalytic performance, we further investigated how sulfur precursor concentration governs morphological evolution during DES–mediated synthesis. FESEM analysis of Co–S nanomaterials synthesized across a $Na_2S_2O_3$ gradient (50–250 mM) revealed distinct structural progression (Fig. 3a–e). The synthesis was conducted via a one–step Ethaline–assisted process in the Ethaline/DES system at 80 °C for 6 h. At 50 mM $Na_2S_2O_3$, irregular nanosheets interconnected, forming a three–dimensional porous network (Fig. 3a). Increasing the concentration to 100 mM induced secondary particle growth (~ 38 nm) on nanosheet surfaces, indicative of sulfur–mediated surface reconstruction (Fig. 3b). At 150 mM, a topological transition occurred, generating monodispersed quasi–spherical nanoparticles of ~ 78 nm in size (Fig. 3c). Notably, at 200 mM, severe

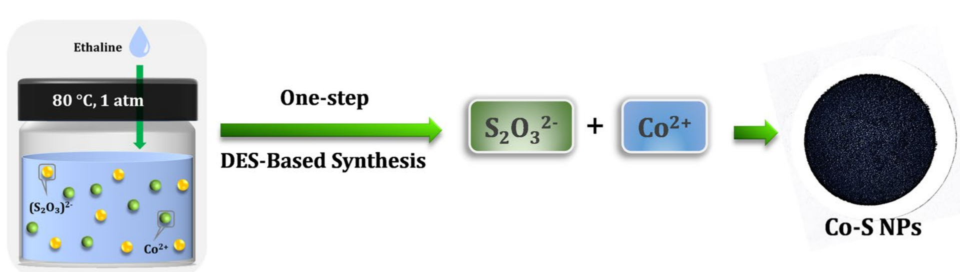


**Fig. 2** Schematic illustration of the one–step Ethaline–assisted synthesis of Co–S nanoparticles (NPs) in the 0.20 M $CoCl_2$–0.15 M $Na_2S_2O_3$–Ethaline system

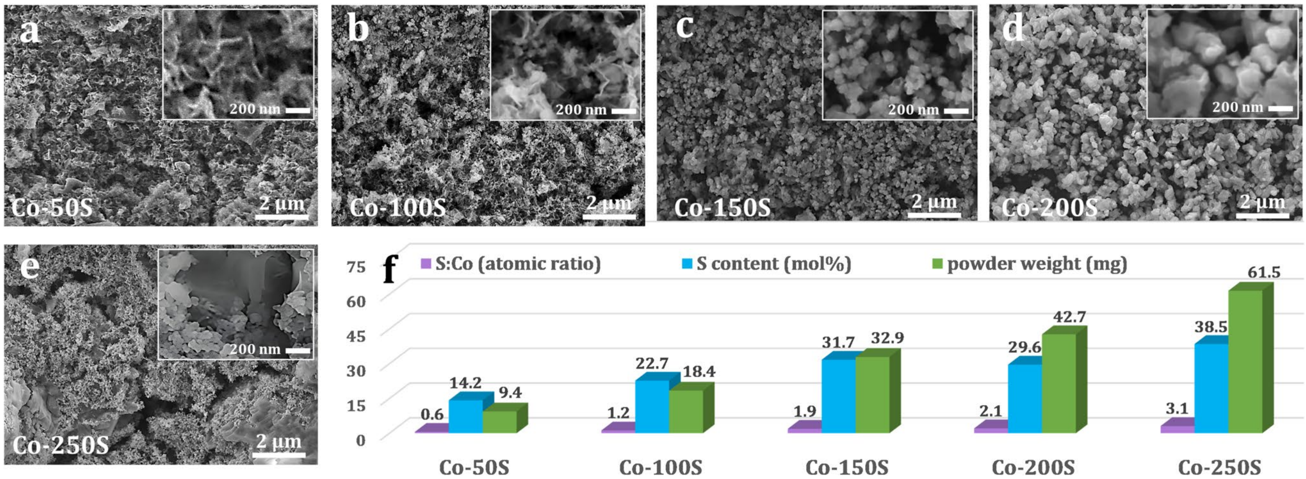


**Fig. 3** (**a**–**e**) FE–SEM images of Co–xS samples synthesized in the 0.20 M $CoCl_2$–x mM $Na_2S_2O_3$–Ethaline system, where x corresponds to $Na_2S_2O_3$ concentrations ranging from 50 to 250 mM. (**f**) The corresponding atomic ratio of S/Co, sulfur molar content (mol%), and powder weight

particle coarsening (~ 180 nm) and aggregation dominated (Fig. 3d). In contrast, 250 mM $Na_2S_2O_3$ generated micron–scale porous clusters decorated with ~ 54 nm secondary particles and surface stress cracks (Fig. 3e). These morphological changes arose from sulfur supersaturation effects (Fig. 3f), which induced lattice distortion and interfacial tension imbalance.

Changes in product mass, monitored synchronously with morphological evolution (green bars in Fig. 3f), exhibited nonlinear growth dynamics. When the sulfur precursor concentration was increased from 50 to 250 mM, the product mass rose from 9.4 mg to 61.5 mg, representing a 6.5–fold increase. However, the mass growth rates across sequential 50 mM concentration increments displayed a non–monotonic decline: 95.7%, 78.8%, 29.8%, and 44%. This nonlinear behavior can be attributed to the concentration–dependent coordination chemistry of sulfur species and their dynamic interactions with the DES solvation environment [41]. At lower concentrations, $S_2O_3^{2-}$ promoted the dissociation of $[CoCl_4]^{2-}$ via chelation, thereby accelerated nucleation, whereas the hydrogen–bonding network of DES suppressed particle aggregation through steric hindrance. Conversely, at higher concentrations, excess $S^{2-}$ delayed nucleation by stabilizing $CoS_x$ polynuclear complexes, while the high viscosity of DES exacerbated diffusion–limited growth, ultimately inducing particle coarsening and aggregation. EDS quantification confirmed that sulfur incorporation efficiency scaled with precursor concentration (cyan bars, Fig. 3f), as evidenced by S/Co atomic ratios increasing from 0.6 (50 mM) to 3.1 (250 mM) (purple bars, Fig. 3f). Notably, the final S/Co ratios consistently surpassed the stoichiometric feed ratios, a phenomenon ascribed to the unique solvation properties of DES. Specifically, the hydroxyl groups of EG in Ethaline formed strong hydrogen bonds with $S^{2-}$, dynamically enriching sulfur species within the local microenvironment and elevating their effective concentration. Furthermore, electrostatic shielding by choline cations ($[Ch]^+$) contributed to the stabilization of Co–S intermediates [38]. These synergistic effects collectively governed the sulfur–dependent structural and dimensional evolution of Co–S, mediated by sulfur's regulatory role in crystal nucleation rates, growth kinetics, and orientation.

To investigate the crystallographic characteristics of Co–S nanoparticles, XRD analyses were conducted. As shown in Fig. 4a, no distinct Bragg diffraction peaks were observed in the XRD pattern of Co–150S NPs (red curve) within the $2\theta$ range of 10°–90°, consistent with an amorphous structure. Intriguingly, elevating the synthesis temperature from 70 °C to 120 °C failed to induce crystallinity in the Co–S samples, in stark contrast to the temperature–driven amorphous–to–crystalline transition reported for Ni–S systems in DES media [43]. This divergence may stem from from either strong coordination interactions between $Co^{2+}$ and $S_2O_3^{2-}$ in the DES solvent that inhibit lattice ordering, or the distinct thermodynamic stabilities of Co and Ni precursors.

Macroscopically, Co–150S NPs appear as a black powder (Fig. 4b), whereas FESEM images (Fig. 4c and d) demonstrate uniformly dispersed nanospheres with a narrow size distribution. EDS quantitative analysis (Fig. 4e) yields a Co/S atomic ratio of 0.53, which closely matches the theoretical stoichiometry of $CoS_2$ (0.5), thereby confirming the formation of Co–S compounds. The detected oxygen signal is attributed to the natural oxide layer on the sample surface. Elemental mapping reveals a homogeneous spatial distribution of Co, S, and O throughout the nanospheres. TEM analysis (Fig. 4f) further

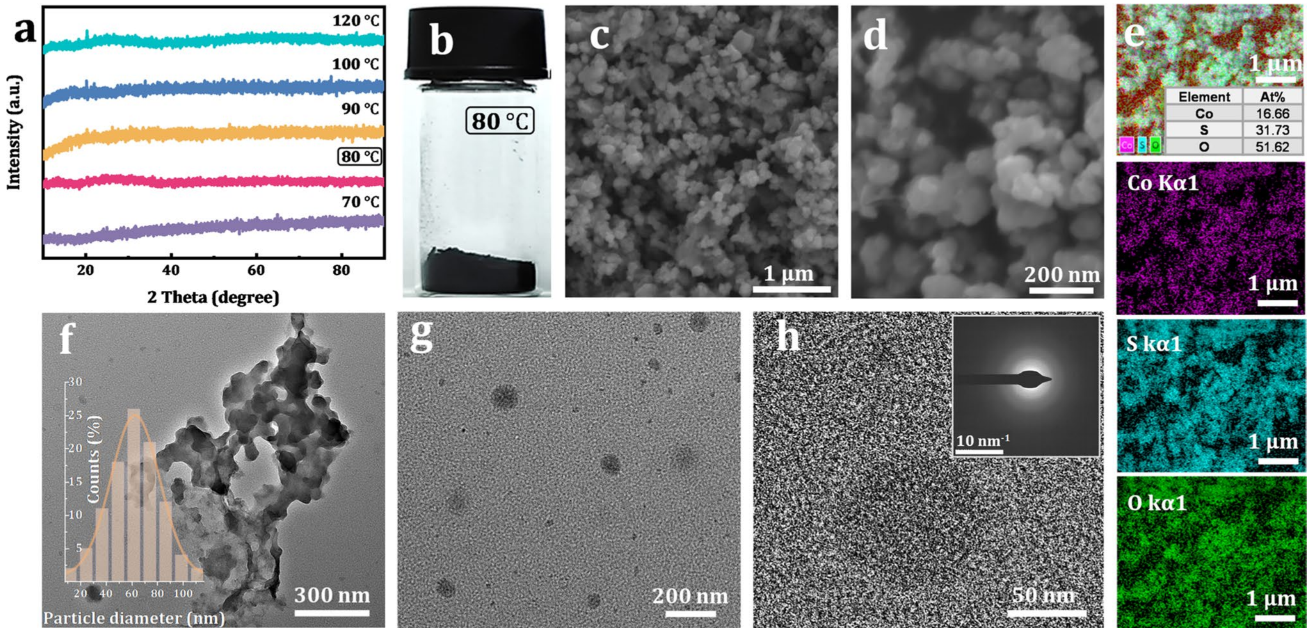


**Fig. 4** (**a**) XRD patterns of Co–S samples synthesized in a 0.20 M $CoCl_2$–0.15 mM $Na_2S_2O_3$–ethylene system at various temperatures. (**b**) Optical image, (**c**–**d**) FESEM images, (**e**) EDS elemental mapping (inset: atomic ratio), and (**f**) TEM image of Co–150S NPs. (**g**–**h**) HRTEM images of Co–150S NPs. Insets: (f) particle size distribution; (h) SAED pattern

corroborates the quasi–spherical morphology, exhibiting an average particle size of ~76 nm, in agreement with SEM observations. HRTEM images (Fig. 4g and h) lack discernible lattice fringes, while the SAED pattern (inset in Fig. 4h) displays no discrete rings or spots. These findings collectively validate the amorphous nature of Co–150S NPs, as evidenced by the XRD results.

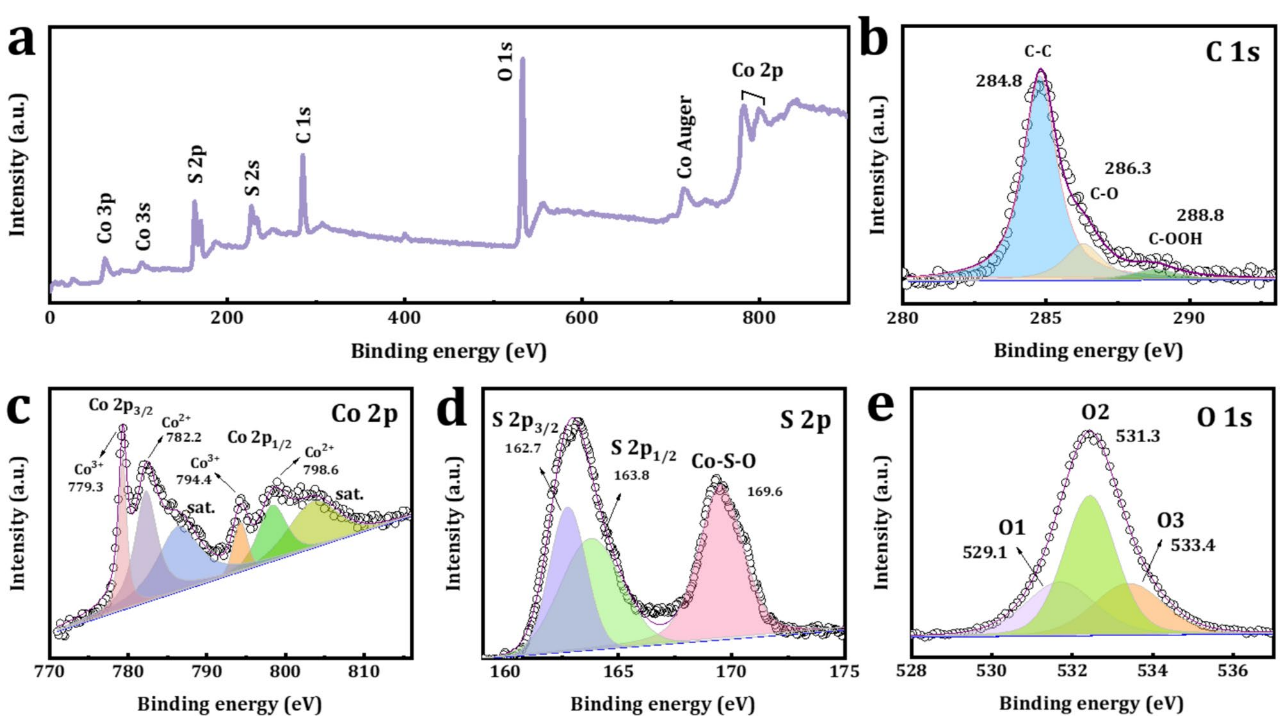


**Fig. 5** High–resolution XPS spectra of Co–150S NPs: (**a**) survey scan, (**b**) C 1 s, (**c**) Co 2p, (**d**) S 2p, and (**e**) O 1 s regions

Figure 5 displays high–resolution XPS spectra of Co–150S NPs. The survey scan (Fig. 5a) confirms the surface composition composed of Co, S, O, and C, consistent with EDS data. The C 1 s spectrum (Fig. 5b) exhibits three characteristic peaks at 284.8, 286.3, and 288.8 eV, assigned to C–C/C = C, C–O, and C = O functional groups characteristic of carbonaceous materials. The dominant peak at 284.8 eV serves as the internal reference for binding energy calibration. The Co 2p spectrum (Fig. 5c) exhibits distinct spin–orbit splitting, where the Co $2p_{3/2}$ and Co $2p_{1/2}$ orbitals split into two sub–peaks. Specifically, the peaks at 779.3 eV (Co $2p_{3/2}$) and 794.2 eV (Co $2p_{1/2}$) are assigned to $Co^{3+}$ species in the Co–S bond, while those at 782.2 eV (Co $2p_{3/2}$) and 798.8 eV (Co $2p_{1/2}$) correspond to $Co^{2+}$ species in Co–O bonds [44, 45].

This dual oxidation state coexistence suggests surface partial oxidation of Co–S phases, attributable to surface oxidation [46]. The S 2p spectrum (Fig. 5d) shows typical doublet peaks at 162.7 eV (S $2p_{3/2}$) and 163.8 eV (S $2p_{1/2}$), with a 1.1 eV splitting consistent with $S^{2-}$ in metal sulfides, confirming Co–S bond formation [47, 48]. The O 1 s spectrum (Fig. 5e) exhibits three components: 529.1 eV (O1) assigned to lattice oxygen (Co–O), 531.3 eV (O2) associated with oxygen vacancies at low–coordination sites, and 533.4 eV (O3) originating from surface–adsorbed oxygen species or physisorbed water [49]. Remarkably, the O2 component constitutes 37.3% of the total oxygen content, indicating a high concentration of surface oxygen vacancies. These defects are likely formed as a result of sulfur atom incorporation during the reaction process, which induces lattice distortion in the cobalt structure. This distortion facilitates the formation of structural defects, including amorphous phases and grain boundaries. Such defects are recognized as active sites for electrocatalytic reactions [15].

Following structural and compositional characterization, the HER performance of Co–xS/NF electrodes synthesized with varying $Na_2S_2O_3$ precursor concentrations (x = 50–250 mM) was systematically evaluated. Linear sweep voltammetry (LSV) was conducted at a scan rate of 5 mV $s^{-1}$ to obtain polarization curves (Fig. 6a). The results showed that the benchmark Pt/C catalyst exhibited exceptional HER activity, whereas the bare NF substrate showed negligible current owing to the absence of active sites. Notably, the HER activity of the Co–xS/NF catalysts depended strongly on sulfur concentration. As the sulfur concentration increased from 50 to 150 mM, the catalytic activity gradually improved. However, further increasing the sulfur concentration beyond 150 mM led to a decline in performance. Among the Co–xS/NF series, Co–150S/NF demonstrated optimal HER activity, requiring an overpotential of only – 105 mV to achieve a

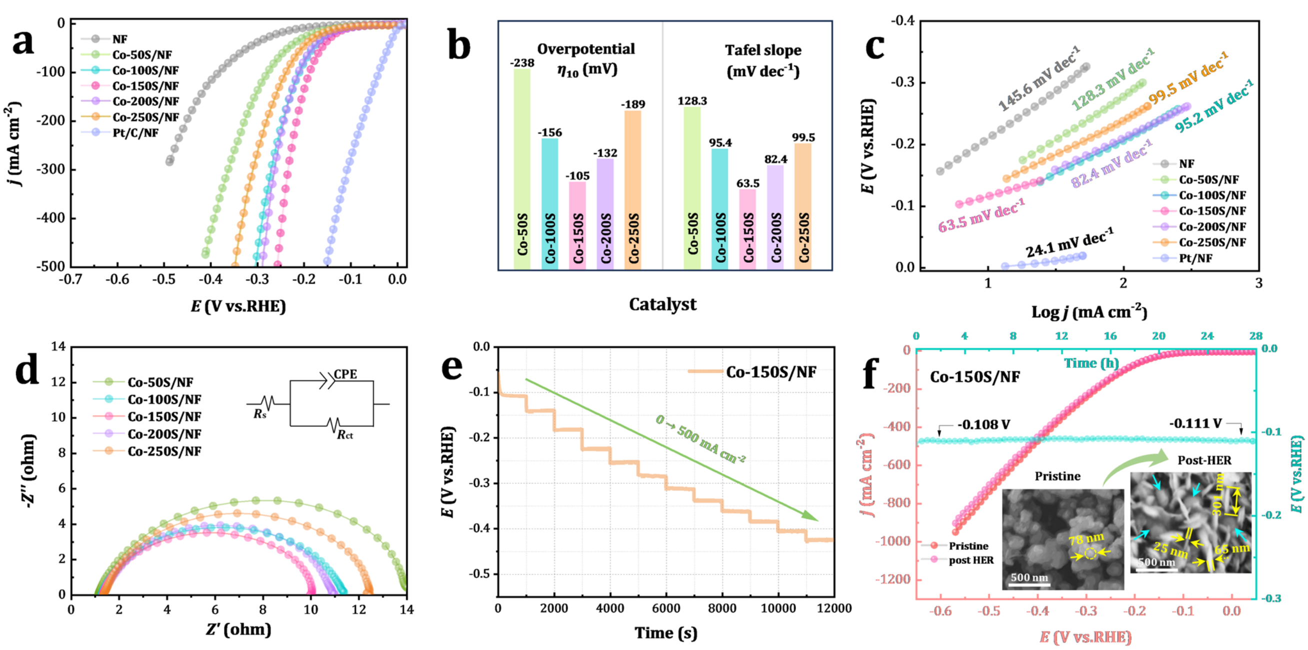


**Fig. 6** (**a**) LSV curves for HER on bare NF, Co–xS/NF, and Pt/C on NF at a scan rate of 5 mV $s^{-1}$. (**b**) Tafel plots with linear fitting for HER of Co–xS/NF. (**c**) Comparison of overpotentials at –10 mA $cm^{-2}$ and Tafel slopes for Co–xS/NF. (**d**) EIS Nyquist plots of Co–xS/NF recorded at an overpotential of –150 mV (vs. RHE). (**e**) Multi–step chronopotentiometric curves of Co–150S/NF without iR compensation, tested sequentially at –10 and –20 mA $cm^{-2}$ (each for 1000 s), followed by incremental steps from –50 to –500 mA $cm^{-2}$ in 50 mA $cm^{-2}$ intervals (each for 1000 s). (**f**) Long–term chronopotentiometric stability test of Co–150S/NF at –10 mA $cm^{-2}$ over 28 h, with LSV curves recorded before and after the stability test. The inset SEM images show the structure changes of post–HER Co–150S/NF after 28 h long–term HER electrolysis at $j$ = –10 mA $cm^{-2}$ in 1.0 M KOH at 25 °C

current density of 10 mA cm$^{-2}$ (Fig. 6b). This corresponds to reductions of 55.9%, 32.7%, 20.6%, and 44.4% compared to Co–50S/NF, Co–100S/NF, Co–200S/NF, and Co–250S/NF, respectively. This observation suggests a "volcano–type" structure–activity relationship, indicating that 150 mM sulfur concentration achieves an optimal balance between active site density and electronic conductivity.

Tafel analysis of the polarization curves (Fig. 6c) yielded a minimum slope of 63.5 mV dec$^{-1}$ for Co–150S/NF, which was significantly lower than those of Co–50S/NF (128.3 mV dec$^{-1}$), Co–100S/NF (95.4 mV dec$^{-1}$), Co–200S/NF (82.4 mV dec$^{-1}$), and Co–250S/NF (99.5 mV dec$^{-1}$) (Fig. 6b). According to classical Tafel theory, this value implies that the HER on Co–150S/NF proceeds via a Volmer–Heyrovsky combined mechanism, with the Heyrovsky step acting as the rate–determining step [50]. Electrochemical impedance spectroscopy (EIS) was performed to investigate the charge transfer kinetics, and Nyquist plots were recorded at − 0.15 V vs. RHE (Fig. 6d). Equivalent circuit fitting of the EIS data revealed that Co–150S/NF exhibited the lowest charge transfer resistance ($R_{ct}$ = 8.7 Ω, Table S1), which was significantly lower than those of Co–50S/NF (12.9 Ω), Co–100S/NF (11.2 Ω), Co–200S/NF (9.7 Ω), and Co–250S/NF (10.2 Ω). To evaluate the structural stability of Co–150S/NF under dynamic current loading, multi–step chronopotentiometric testing was conducted (Fig. 6e). When the current density was incrementally increased from − 10 mA cm$^{-2}$ to − 500 mA cm$^{-2}$ (in 50 mA cm$^{-2}$ steps, each maintained for 1000 s), the electrode potential responded rapidly and remained stable at each step. This rapid response suggests robust structural integrity and efficient bubble detachment capability. Furthermore, long–term stability was further assessed through a 28 h continuous HER test at − 10 mA cm$^{-2}$ in 1.0 M KOH (Fig. 6f). The overpotential showed minimal variation, shifting slightly from − 108 mV to − 111 mV (≈2.8% decay), while the post–HER LSV curves closely overlapped with the initial scan, confirming exceptional durability in alkaline media. Interestingly, characterization of the post–HER sample revealed substantial structural reorganization (Figure S10), despite its moderate HER activity. The morphological evolution exhibited dual distinct features: (1) a phase transition from quasi–spherical nanoparticles to vertically aligned thin nanoflakes (thickness: 25–65 nm; diameter: ~ 301 nm), and (2) the persistence of sparse residual nanospheres embedded within the nanoflake matrix (inset in Fig. 6f). Elemental mapping analysis quantified a striking decrease in S content, which plummeted from 31.7 at% to 6.0 at%, while Co concentrations surged from 16.7 at% to 37.9 at%, with O levels maintaining stability (51.6 at% → 56.1 at%; Figure S10). This pronounced S depletion (81.2% loss), coupled with Co enrichment, suggests an electrochemical dissolution–reorganization process. The resultant Co–rich phase, characterized by disordered atomic arrangements and metastable coordination environments, likely serves as the electrochemically active component. This dynamic structural evolution likely originates from defect–rich interfaces generated during Ethaline–mediated synthesis, where the DES high ionic polarizability promotes kinetically trapped metastable states that facilitate atomic rearrangement.

Analogous to HER trends, the OER activity of Co–xS/NF electrodes followed a sulfur concentration–dependent "volcano–type" relationship (Fig. 7a). Gradually increasing the sulfur precursor concentration from 50 to 150 mM resulted in a continuous improvement in OER kinetics, as evidenced by a decrease in the overpotential at 100 mA cm$^{-2}$ from 378 mV (Co–50S/NF) to 334 mV (Co–150S/NF) (Fig. 7b). This trend underscores the synergistic role of sulfur in enhancing charge transfer and exposing more active sites. However, when the precursor concentration exceeded the optimal 150 mM, OER performance paradoxically deteriorated. Co–200S/NF (200 mM) and Co–250S/NF (250 mM) exhibited higher overpotentials of 342 mV and 372 mV, respectively. This reversal suggests a threshold effect, where excessive sulfur precursor likely induces phase segregation and/or obstructs electrochemical interfaces due to competing surface adsorption processes during electrode synthesis.

Kinetic analysis (Fig. 7c) identified Co–150S/NF's minimal Tafel slope (73.2 mV dec$^{-1}$), 38.6% and 18.8% lower than Co–50S/NF (119.4 mV dec$^{-1}$) and Co–250S/NF (90.2 mV dec$^{-1}$), consistent with an adsorbate evolution mechanism [51]. EIS at 350 mV overpotential (Fig. 7d) corroborated these results, showing Co–150S/NF's charge transfer resistance ($R_{ct}$ = 2.6 Ω) was 54.4% smaller than Co–50S/NF (5.7 Ω), reflecting optimized electronic structure and abundant active sites. To further probe the electrochemically active surface area (ECSA), double-layer capacitance ($C_{dl}$) measurements were conducted via cyclic voltammetry in the non-Faradaic region (0.93–1.03 V vs. RHE). As shown in Figure S11, Co–150S/NF exhibited the highest $C_{dl}$ (6.7 mF cm$^{-2}$) and ECSA (3.7 cm$^{2}$), surpassing other samples ($C_{dl}$: 2.8–5.9 mF cm$^{-2}$; ECSA: 1.6–3.3 cm$^{2}$), consistent with a "volcano-like" dependence on sulfur content. This optimal sulfur doping enhanced active site exposure through its dispersed quasi-spherical morphology, amorphous defects, and sulfur-mediated charge redistribution. Operational stability testing (Fig. 7e) confirmed Co–150S/NF's robustness, as potentials remained stable during dynamic current loading (10 → 500 mA cm$^{-2}$) and post–6000 CV cycle polarization curves exhibited negligible deviation. Long–term electrolysis at 10 mA cm$^{-2}$ for 28 h (Fig. 7f) exhibited a negligible overpotential increase from 1.507 V to 1.511 V ($\Delta\eta$ = 4 mV), demonstrating exceptional OER stability in alkaline media. Notably, post–test characterization revealed a complete morphological transformation of Co–150S/NF into oxygen–enriched ultrathin nanosheets with thicknesses

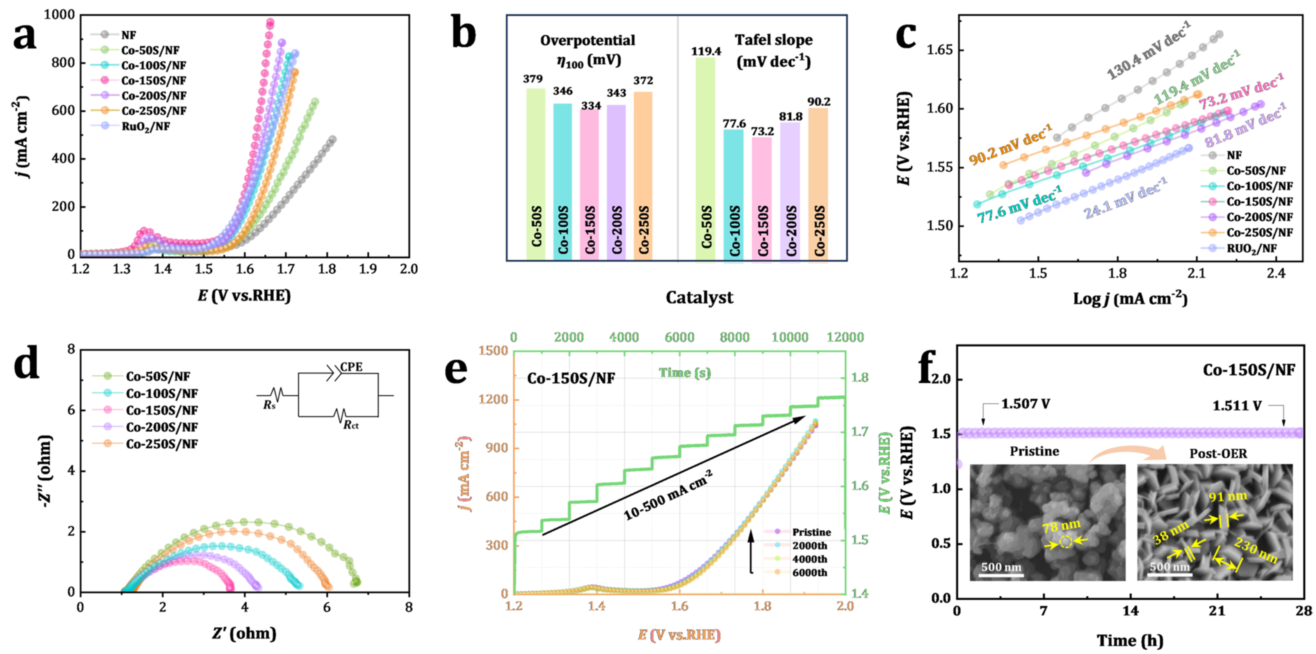


**Fig. 7** (**a**) LSV curves for OER on bare NF, Co–xS/NF, and $RuO_2$/NF at a scan rate of 5 mV $s^{-1}$. (**b**) Comparison of overpotentials at 100 mA $cm^{-2}$ and Tafel slopes for Co–xS/NF. (**d**) EIS Nyquist plots for Co–xS/NF recorded at an overpotential of 350 mV (vs. RHE). (**c**) Tafel plots with linear fitting for OER of Co–xS/NF. (**e**) Multi–step chronopotentiometric curves for Co–150S/NF without iR compensation, tested sequentially at 10 and 20 mA $cm^{-2}$ (each for 1000 s), followed by incremental steps from 50 to 500 mA $cm^{-2}$ in 50 mA $cm^{-2}$ intervals (each for 1000 s), and LSV curves of Co–S–50/NF before and after 2000, 4000, and 6000 CV cycles (data without iR correction). (**f**) Long–term chronopotentiometric stability test of Co–150S/NF at 10 mA $cm^{-2}$ over 28 h. The inset SEM images show the structure changes of post–OER Co–150S/NF after 28 h long–term OER electrolysis at $j = 10$ mA $cm^{-2}$ in 1.0 M KOH at 25 °C

of 38–91 nm and lateral dimensions of ~230 nm (inset in Fig. 7f). This transformation was driven by near–complete sulfur depletion (87.0% loss) and concomitant oxygen accumulation (59.5% gain), while cobalt content remained stable (16.7 at% → 13.6 at%) (Figure S12). The reconstruction mechanism involves two synergistic processes: (i) electrochemical sulfur oxidation driving the phase transition from Co–S to Co–O/OH, and (ii) oxygen evolution–derived microbubbles generating interfacial shear forces that template nanosheet alignment and mesopore formation [52]. The self–optimized Co–O/OH architecture integrates undercoordinated active sites with hierarchically porous nanosheet arrays, which collectively resolve the activity–stability trade–off through synergistic coordination of catalytic turnover kinetics and bubble–enhanced mass transport.

Building on the dual–functional catalytic advantages of Co–150S/NF in both HER and OER, a dual–electrode overall water splitting system was constructed using Co–150S/NF as both the anode and cathode. In a 1.0 M KOH electrolyte, the system required only 1.82 V to drive a current density of 100 mA $cm^{-2}$ (Fig. 8a), outperforming the $RuO_2$/NF||Pt/C/NF benchmark (1.85 V) and bare NF (2.08 V). The evolution of hydrogen and oxygen gases was measured during electrolysis at 20 mA $cm^{-2}$ (Fig. 8b). The amounts of experimentally measured gases matched well with theoretically calculated values, and the molar ratio of produced hydrogen and oxygen was close to 2:1, indicating that the Faradaic efficiency approached ~100% for both HER and OER, with the ratio of $H_2$ to $O_2$ being close to 2:1. Long–term stability testing (Fig. 8c) showed that after 26 h of continuous operation at a constant current density of 10 mA $cm^{-2}$, the cell voltage increased from 1.62 V to 1.66 V ($\Delta V = 40$ mV), with an average degradation rate of only 1.5 mV $h^{-1}$. The remarkable catalytic performance of Co–150S/NF, coupled with its simple and scalable synthesis process, makes it a promising candidate for practical water electrolysis devices.

The exceptional bifunctional electrocatalytic performance of Co–150S NPs in overall water splitting arises from multiscale synergistic effects induced by sulfur incorporation, driven by three interconnected mechanisms: (1) Morphological Control: The dynamic hydrogen–bond network of the non–aqueous DES solvent, combined with coordinated $S_2O_3^{2-}$ precursors, synergistically regulates nucleation kinetics. This suppresses particle aggregation through spatial confinement, resulting in well–dispersed quasi–spherical nanoparticles. Their low surface curvature facilitates electrolyte diffusion and bubble detachment; (2) Electronic Structure Optimization: Sulfur incorporation promotes the formation of non–stoichiometric Co–S composites, which optimize charge transfer efficiency via tailored electronic

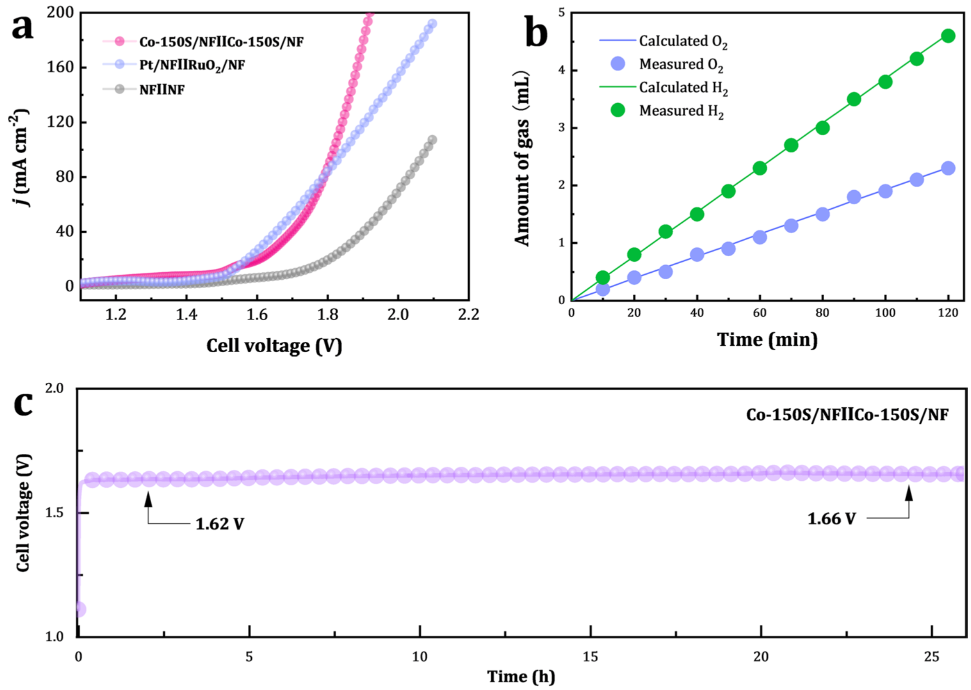


**Fig. 8** (**a**) LSV curves for overall water splitting using coupled Co–150S/NF, bare NF, and $RuO_2$/NF||Pt/C/NF at a scan rate of 5 mV $s^{-1}$. (**b**) Comparison of theoretical and experimentally measured volumes of $H_2$ and $O_2$ evolved on Co–150S/NF during overall water splitting at 20 mA $cm^{-2}$. (**c**) Long–term chronopotentiometric stability test of Co–150S/NF for overall water splitting at 10 mA $cm^{-2}$ over 26 h

configurations; (3) Defect Activation: The amorphous nanostructure, coupled with oxygen vacancies, exposes defect–rich regions that serve as dual–active centers, simultaneously lowering the energy barriers for both HER and OER. This morphology–electronic–defect synergy enables Co–150S NPs to surpass noble metal benchmarks in both activity and stability, offering a novel strategy for designing cost–effective water–splitting catalysts.

## Conclusions

This study demonstrates the low–temperature controllable synthesis of amorphous Co–S bifunctional electrocatalysts via a green Ethaline/DES–driven strategy. By optimizing the dynamic coordination between $Co^{2+}$ and $S_2O_3^{2-}$ in the 0.20 M $CoCl_2$–0.15 M $Na_2S_2O_3$–Ethaline system (80 °C, 6 h), monodisperse quasi-spherical Co–150S NPs (~ 78 nm) were synthesized with tailored sulfur content (S/Co = 1.9), amorphous architecture, and oxygen vacancy enrichment. Electrochemical analysis revealed a volcano-type sulfur-dependent activity profile, with Co–150S/NF achieving the highest $C_{dl}$ (6.7 mF $cm^{-2}$) and ECSA (3.7 $cm^2$), confirming optimal active site exposure. These structural merits synergistically enhanced charge transfer kinetics and active site density, enabling Co–150S/NF to deliver exceptional bifunctional performance in 1 M KOH: HER overpotential of − 105 mV at 10 mA $cm^{-2}$ (Tafel slope: 63.5 mV $dec^{-1}$) and OER overpotential of 277 mV at 10 mA $cm^{-2}$ (Tafel slope: 73.2 mV $dec^{-1}$), outperforming low–sulfur counterparts while achieving comparable performance to $RuO_2$ and Pt/C. For overall water splitting, the catalyst requires only 1.62 V to drive 10 mA $cm^{-2}$, with a degradation rate of 1.5 mV $h^{-1}$ over 26 h. Mechanistic studies reveal that sulfur incorporation enhances the catalytic performance through a multiscale synergistic mechanism: morphological regulation (spherical dispersion suppresses aggregation), electronic structure optimization (non–stoichiometric Co–S complexes enhance conductivity), and defect enhancement (oxygen vacancy–rich regions lower the reaction energy barriers). This work establishes a new paradigm for developing efficient, low–cost water–splitting catalysts and highlights the unique advantages of DES–based systems in precision nanomaterial synthesis.

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**Author contributions** Youpo Mise, Shaohua Wang, Wen Shi, and Yakun Yin carried out experiments, Wenqiang Yang conceived the project, Wenqiang Yang, Wentang Xia, Xuejiao Zhou and Juan An supervised electric cell and electrochemical aspects of the project, and all authors cowrote the manuscript.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare no competing interests.

## References


1. Xu Z, Liu Y, Cheng X, Shang Y, Wang Y, Wang J, Li W, He G (2025) A catalyst-coated diaphragm assembly to improve the performance and energy efficiency of alkaline water electrolysers. Commun Eng 4(1):9. https://doi.org/10.1038/s44172-025-00344-2
2. Zou X, Zhang Y (2015) Noble metal-free hydrogen evolution catalysts for water splitting. Chem Soc Rev 44:5148–5180. https://doi.org/10.1039/c4cs00448e
3. Cao X, Wang T, Jiao LJAFM (2021) Transition-metal (Fe Co, and Ni)-based nanofiber electrocatalysts for water splitting. Adv Fiber Mater 3:210–228. https://doi.org/10.1007/s42765-021-00065-z
4. Sun W, Zhu J, Zhang M, Meng X, Chen M, Feng Y, Chen X, Ding Y (2022) Recent advances and perspectives in cobalt-based heterogeneous catalysts for photocatalytic water splitting, CO2 reduction, and N2 fixation. Chin J Catal 43:2273–2300. https://doi.org/10.1016/s1872-2067(21)63939-6
5. Wang J, Cui W, Liu Q, Xing Z, Asiri AM, Sun X (2016) Recent progress in cobalt-based heterogeneous catalysts for electrochemical water splitting. Adv Mater 28:215–230. https://doi.org/10.1002/adma.201502696
6. Huang C, Qin P, Luo Y, Ruan Q, Liu L, Wu Y, Li Q, Xu Y, Liu R, Chu PKJMTE (2022) Recent progress and perspective of cobalt-based catalysts for water splitting: design and nanoarchitectonics. Mater Today Energy 23:100911. https://doi.org/10.1016/j.mtener.2021.100911
7. Guan C, Liu X, Elshahawy AM, Zhang H, Wu H, Pennycook SJ, Wang J (2017) Metal–organic framework derived hollow CoS2 nanotube arrays: an efficient bifunctional electrocatalyst for overall water splitting. Nanoscale Horiz 2:342–348. https://doi.org/10.1039/c7nh00079k
8. Wei S, Wang X, Wang J, Sun X, Cui L, Yang W, Zheng Y, Liu JJEA (2017) CoS2 nanoneedle array on Ti mesh: A stable and efficient bifunctional electrocatalyst for urea-assisted electrolytic hydrogen production. Electrochim Acta 246:776–782. https://doi.org/10.1016/j.electacta.2017.06.068
9. Ahn I-K, Joo W, Lee J-H, Kim HG, Lee S-Y, Jung Y, Kim J-Y, Lee G-B, Kim M, Joo Y-C (2019) Metal-organic framework-driven porous cobalt disulfide nanoparticles fabricated by gaseous sulfurization as bifunctional electrocatalysts for overall water splitting. Sci Rep 9:19539. https://doi.org/10.1038/s41598-019-56084-9
10. Ma X, Zhang W, Deng Y, Zhong C, Hu W, Han XJN (2018) Phase and composition controlled synthesis of cobalt sulfide hollow nanospheres for electrocatalytic water splitting. Nanoscale 10:4816–4824. https://doi.org/10.1039/c7nr09424h
11. Yang Y, Yao H, Yu Z, Islam SM, He H, Yuan M, Yue Y, Xu K, Hao W, Sun GJ (2019) Hierarchical nanoassembly of MoS2/Co9S8/Ni3S2/Ni as a highly efficient electrocatalyst for overall water splitting in a wide pH range. J Am Chem Soc 141:10417–10430. https://doi.org/10.1021/jacs.9b04492
12. Feng Z, Pu J, Liu M, Zhang W, Zhang X, Cui L, Liu J (2022) Facile construction of hierarchical Co3S4/CeO2 heterogeneous nanorod array on cobalt foam for electrocatalytic overall water splitting. J Colloid Interface Sci 613:806–813. https://doi.org/10.1016/j.jcis.2022.01.081
13. Dong Y, Fang Z, Yang W, Tang B, Liu QJAAM (2022) Integrated bifunctional electrodes based on amorphous Co–Ni–S nanoflake arrays with atomic dispersity of active sites for overall water splitting. ACS Appl Mater Interfaces 14:10277–10287. https://doi.org/10.1021/acsami.1c22092
14. Kang Z, Guo H, Wu J, Sun X, Zhang Z, Liao Q, Zhang S, Si H, Wu P, Wang LJAFM (2019) Engineering an earth-abundant element-based bifunctional electrocatalyst for highly efficient and durable overall water splitting. Adv Funct Mater 29:1807031. https://doi.org/10.1002/adfm.201807031
15. Yang W, Zeng J, Hua Y, Xu C, Siwal SS, Zhang Q (2019) Defect engineering of cobalt microspheres by S doping and electrochemical oxidation as efficient bifunctional and durable electrocatalysts for water splitting at high current densities. J Power Sources 436:226887. https://doi.org/10.1016/j.jpowsour.2019.226887
16. Zhu H, Zhang J, Yanzhang R, Du M, Wang Q, Gao G, Wu J, Wu G, Zhang M, Liu B (2015) When cubic cobalt sulfide meets layered molybdenum disulfide: a core-shell system toward synergetic electrocatalytic water splitting. Adv Mater 27:4752–4759. https://doi.org/10.1002/adma.201501969
17. Shen J, Liu Y, Chen Q, Yu W, Zhong Q (2024) In-situ construction defect-rich CuNiCoS4/1T-MoS2 heterostructures as superior electrocatalysts for water splitting. J Colloid Interface Sci 658:1009–1015. https://doi.org/10.1016/j.jcis.2023.12.027
18. Singh S, Irshad A, De la Cruz GD, Zhao B, Zayat B, Kwon Y, Chang Q, Narayan SR, Ravichandran J (2025) Bifunctional noble metal-free ternary chalcogenide electrocatalysts for overall water splitting. Chem Mat 37:823–832. https://doi.org/10.1021/acs.chemmater.4c01684
19. Wang C, Du X, Zhang X (2023) Controlled synthesis of W-Co3S4@Co3O4 as an environmentally friendly and low cost electrocatalyst for overall water splitting. Int J Hydrogen Energy 48:12739–12752. https://doi.org/10.1016/j.ijhydene.2022.12.241
20. Huang Y, Jiang L-W, Liu H, Wang J-J (2022) Electronic structure regulation and polysulfide bonding of Co-doped (Ni, Fe)1+xS enable highly efficient and stable electrocatalytic overall water splitting. Chem Eng J 441:136121. https://doi.org/10.1016/j.cej.2022.136121
21. Su H, Song S, Li S, Gao Y, Ge L, Song W, Ma T, Liu J (2021) High-valent bimetal Ni3S2/Co3S4 induced by Cu doping for bifunctional electrocatalytic water splitting. Appl Catal B Environ 293:120225. https://doi.org/10.1016/j.apcatb.2021.120225
22. Liu Y, Li Q, Si R, Li G-D, Li W, Liu D-P, Wang D, Sun L, Zhang Y, Zou X (2017) Coupling sub-nanometric copper clusters with quasi-amorphous cobalt sulfide yields efficient and robust electrocatalysts for water splitting reaction. Adv Mater 29:1606200. https://doi.org/10.1002/adma.201606200
23. Zhang C, Zhang B, Li Z, Hao J (2019) Deep eutectic solvent-mediated hierarchically structured Fe-based organic–inorganic hybrid catalyst for oxygen evolution reaction. ACS Appl Energy Mater 2:3343–3351. https://doi.org/10.1021/acsaem.9b00197
24. Jiang J, Chang L, Zhao W, Tian Q, Xu Q (2019) An advanced FeCoNi nitro-sulfide hierarchical structure from deep eutectic solvents for enhanced oxygen evolution reaction. Chem Commun 55:10174–10177. https://doi.org/10.1039/c9cc05389a
25. Hartley JM, Ip C-M, Forrest GC, Singh K, Gurman SJ, Ryder KS, Abbott AP, Frisch G (2014) EXAFS study into the speciation of metal salts dissolved in ionic liquids and deep eutectic solvents. Inorg Chem 53:6280–6288. https://doi.org/10.1021/ic500824r
26. Dag Ö, Alayoğlu S, Uysal İ (2004) Effects of ions on the liquid crystalline mesophase of transition-metal salt: Surfactant (CnEOm). J Phys Chem B 108:8439–8446. https://doi.org/10.1021/jp049716x
27. Gallo V, Mastrorilli P, Nobile CF, Romanazzi G, Suranna GP (2002) How does the presence of impurities change the performance of catalytic systems in ionic liquids? A case study: the Michael addition of acetylacetone to methyl vinyl ketone. J Chem Soc Dalton Trans 2:4339–4342. https://doi.org/10.1039/b207526a

28. Li A, Ren S, Teng C, Liu H, Zhang Q (2022) The role of deep eutectic solvents in chiral capillary electrokinetic chromatography: a comparative study based on $\alpha$-cyclodextrin chiral selector. J Mol Liq 359:119281. https://doi.org/10.1016/j.molliq.2022.119281
29. Jurić T, Uka D, Holló BB, Jović B, Kordić B, Popović BM (2021) Comprehensive physicochemical evaluation of choline chloride-based natural deep eutectic solvents. J Mol Liq 343:116968. https://doi.org/10.1016/j.molliq.2021.116968
30. Chai K, Lu X, Zhou Y, Liu H, Wang G, Jing Z, Zhu F, Han L (2022) Hydrogen bonds in aqueous choline chloride solutions by DFT calculations and X-ray scattering. J Mol Liq 362(1):19742. https://doi.org/10.1016/j.molliq.2022.119742
31. Stefanescu M, Stoia M, Stefanescu O (2007) Thermal and FT-IR study of the hybrid ethylene-glycol–silica matrix. J Sol-Gel Sci Technol 41:71–78. https://doi.org/10.1007/s10971-006-0118-5
32. Chen L, Chao Y, Li X, Zhou G, Lu Q, Hua M, Li H, Ni X, Wu P, Zhu WJGC (2021) Engineering a tandem leaching system for the highly selective recycling of valuable metals from spent Li-ion batteries. Green Chem 23:2177–2184. https://doi.org/10.1039/d0gc03820b
33. Cao X, Xu L, Shi Y, Wang Y, Xue XJEA (2019) Electrochemical behavior and electrodeposition of cobalt from choline chloride-urea deep eutectic solvent. Electrochim Acta 295:550–557. https://doi.org/10.1016/j.electacta.2018.10.163
34. Wang S, Zhang Z, Lu Z, Xu ZJGC (2020) A novel method for screening deep eutectic solvent to recycle the cathode of Li-ion batteries. Green Chem 22:4473–4482. https://doi.org/10.1039/d0gc00701c
35. Lu Q, Chen L, Li X, Chao Y, Sun J, Ji H, Zhu WJASC (2021) Engineering, Sustainable and convenient recovery of valuable metals from spent Li-ion batteries by a one-pot extraction process. ACS Sustain Chem Eng 9:13851–13861. https://doi.org/10.1021/acssuschemeng.1c04717
36. Chen W, Xue Z, Wang J, Jiang J, Zhao X, Mu TJAPCS (2018) Investigation on the thermal stability of deep eutectic solvents. Acta Phys Chim Sin 34:904–911. https://doi.org/10.3866/PKU.WHXB201712281
37. Hansen BB, Spittle S, Chen B, Poe D, Sangoro JRJCR (2020) Deep eutectic solvents: a review of fundamentals and applications. Chem Rev 121:1232–1285. https://doi.org/10.1021/acs.chemrev.0c00385
38. Abbott AP, Capper G, Davies DL, Rasheed RK, Tambyrajah V (2003) Novel solvent properties of choline chloride/urea mixtures. Chem Commun 1:70–71. https://doi.org/10.1039/B210714G
39. D'Agostino C, Harris RC, Abbott AP, Gladden LF, Mantle MD (2011) Molecular motion and ion diffusion in choline chloride based deep eutectic solvents studied by 1 H pulsed field gradient NMR spectroscopy. Phys Chem Chem Phys 13:21383–21391. https://doi.org/10.1039/c1cp22554e
40. Yue D, Jia Y, Yao Y, Sun J, Jing Y (2012) Structure and electrochemical behavior of ionic liquid analogue based on choline chloride and urea. Electrochim Acta 65:30–36. https://doi.org/10.1016/j.electacta.2012.01.003
41. Smith EL, Abbott AP, Ryder KS (2014) Deep eutectic solvents (DESs) and their applications. Chem Rev 114:11060–11082. https://doi.org/10.1021/cr300162p
42. Atilhan M, Aparicio S (2017) behavior of deep eutectic solvents under external electric fields: a molecular dynamics approach. J Phys Chem B 121:221–232. https://doi.org/10.1021/acs.jpcb.6b09714
43. Zhang Y, Ru J, Hua Y, Huang P, Bu J, Wang Z (2021) Facile and controllable synthesis of NiS2 nanospheres in deep eutectic solvent. Mater Lett 283:128742. https://doi.org/10.1016/j.matlet.2020.128742
44. Hua Y, Jiang H, Jiang H, Zhang H, Li C (2018) Hierarchical porous CoS2 microboxes for efficient oxygen evolution reaction. Electrochim Acta 278:219–225. https://doi.org/10.1016/j.electacta.2018.05.028
45. Li Y, Wang Z, Hu J, Li S, Du Y, Han X, Xu P (2020) Metal–organic frameworks derived interconnected bimetallic metaphosphate nanoarrays for efficient electrocatalytic oxygen evolution. Adv Funct Mater 30:1910498. https://doi.org/10.1002/adfm.201910498
46. Xie Z, Tang H, Wang Y (2019) MOF-derived Ni-doped CoS2 grown on carbon fiber paper for efficient oxygen evolution reaction. ChemElectroChem 6:1206–1212. https://doi.org/10.1002/celc.201801106
47. Zhu H, Zhang J, Yanzhang R, Du M, Wang Q, Gao G, Wu J, Wu G, Zhang M, Liu B (2015) When cubic cobalt sulfide meets layered molybdenum disulfide: a core-shell system toward synergetic electrocatalytic water splitting. Adv Mater 27:4752–4759. https://doi.org/10.1002/adma.201501969
48. Wang C, Wang T, Liu J, Zhou Y, Yu D, Han F, Li Q, Chen J, Huang Y (2018) Facile synthesis of silk-cocoon S-rich cobalt polysulfide as an efficient catalyst for the hydrogen evolution reaction. Energy Environ Sci 11:2467–2475. https://doi.org/10.1039/c8ee00948a
49. Yang WQ, Hua YX, Zhang QB, Lei H, Xu CY (2018) Electrochemical fabrication of 3D quasi-amorphous pompon-like Co-O and Co-Se hybrid films from choline chloride/urea deep eutectic solvent for efficient overall water splitting. Electrochim Acta 273:71–79. https://doi.org/10.1016/j.electacta.2018.04.054
50. Zheng Y, Jiao Y, Jaroniec M, Qiao SZ (2015) Advancing the electrochemistry of the hydrogen-evolution reaction through combining experiment and theory. Angew Chem Int Ed 54:52–65. https://doi.org/10.1002/anie.201407031
51. Suen N-T, Hung S-F, Quan Q, Zhang N, Xu Y-J, Chen HM (2017) Electrocatalysis for the oxygen evolution reaction: recent development and future perspectives. Chem Soc Rev 46:337–365. https://doi.org/10.1039/c6cs00328a
52. Yang C, Gao MY, Zhang QB, Zeng JR, Li XT, Abbott AP (2017) In-situ activation of self-supported 3D hierarchically porous Ni3S2 films grown on nanoporous copper as excellent pH-universal electrocatalysts for hydrogen evolution reaction. Nano Energy 36:85–94. https://doi.org/10.1016/j.nanoen.2017.04.032